

Supplementary Material

A Features Importance

In this section, we explore the effect of individual features on the overall performance of ORE. Recall from Section 4, the hybrid setting uses $BM(q, d)$ (or x_1), $TCT(q, d)$ (or x_2), $RM3(q', d)$ (or x_3), and $D2SETAFF$ (a combination of $RM3(q', d)$ i.e., x_3 and $TCT(d, d')$ i.e., x_5). Since, we learn $\vec{\alpha}$ per query level, using these values as features' importance might not give insight for queries in the test set. Therefore, we chose to show the importance of the features by observing the average performance drop after removing certain features. Figure 6, we report the performance of ORE by removing the features

one by one (with replacement). We observe that the removal of the feature $TCT(q, d)$ (or x_2), which is denoted by w/o TCT , shows the highest drop in performance. Recall@50 drops from 0.558 to 0.473. Similarly, nDCG@50 drops from 0.683 to 0.617.

In adaptive setting, ORE uses the $Q2DAFF(q, d) = BM25(q, d)$, $D2SETAFF(d, S)$ (defined in Equation 4), and feature x_7 (defined in Equation 5). We report the performance for the adaptive retrieval setting in Figure 7. The feature x_7 seems to be the most prominent (denoted by w/o x_7). Recall@50 drops from 0.509 to 0.463, and nDCG@50 drops from 0.640 to 0.594. While at budget 100, we do not observe huge performance drops in ranking and retrieval.

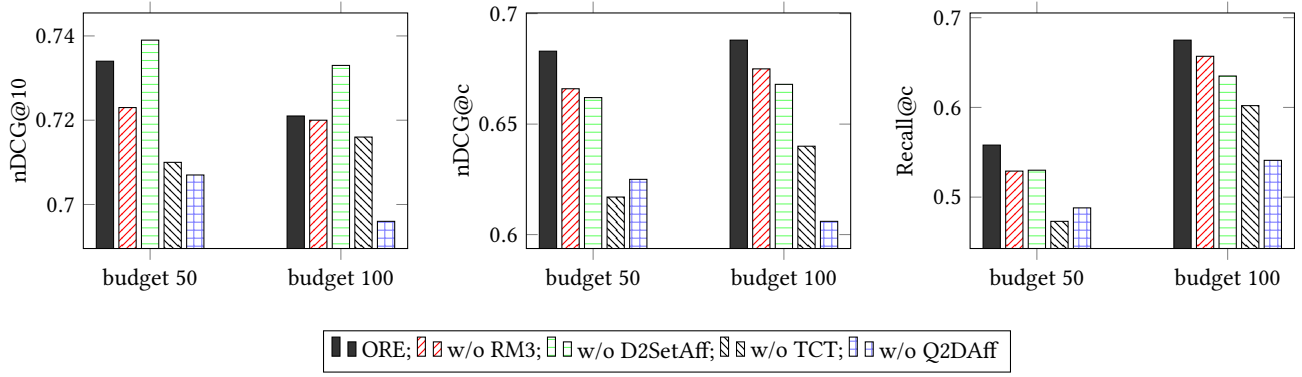


Figure 6: Effect of different features on ORE's performance in Hybrid retrieval setting.

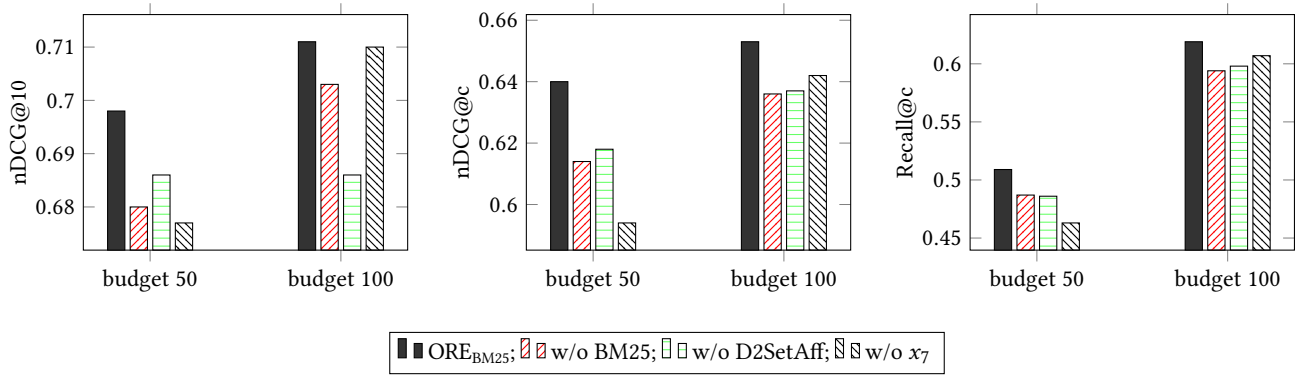


Figure 7: Effect of different features on ORE's performance in Adaptive retrieval setting.